



Gross and Microscopic Pathology

Image review session —
Lecture 9

Department of Pathology



Wet and dry gangrene

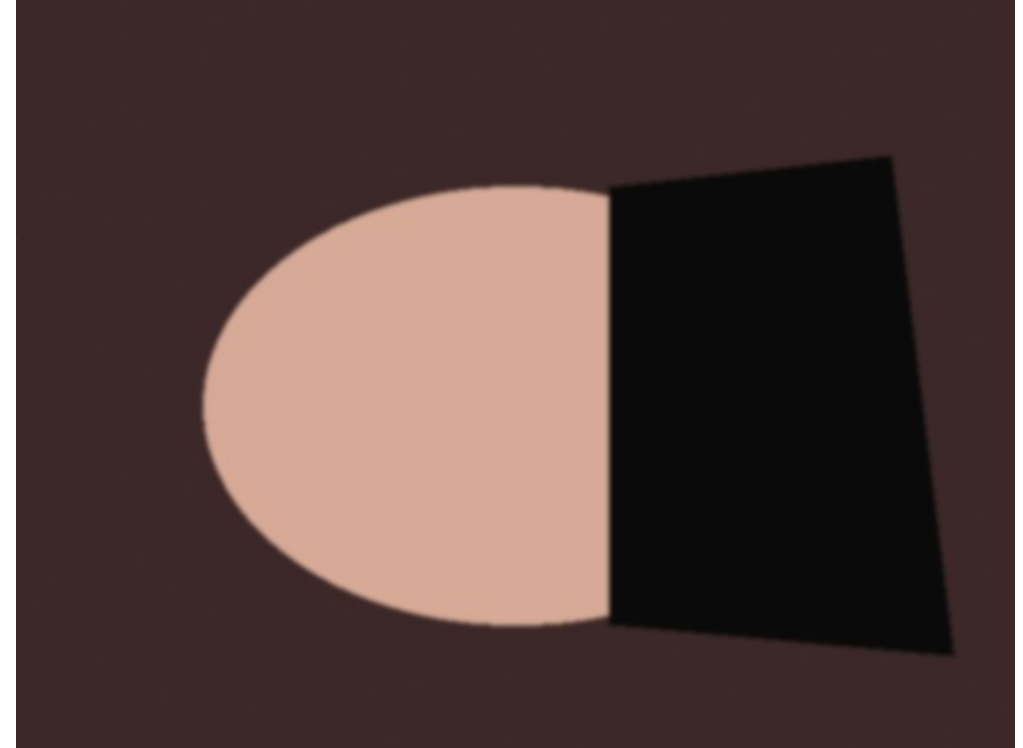
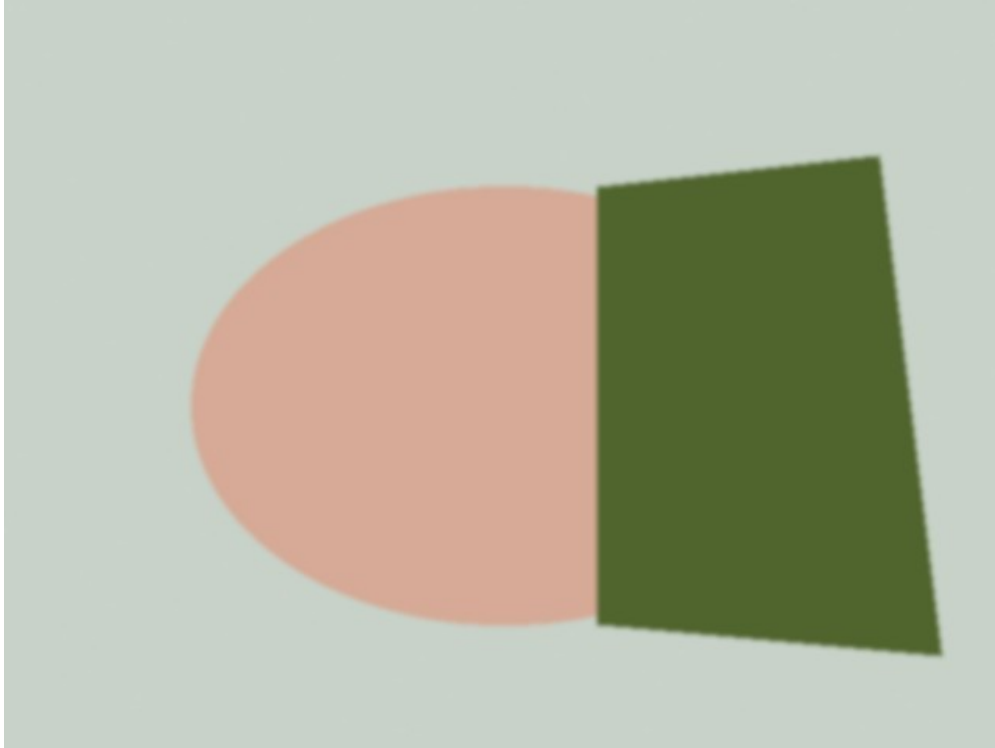
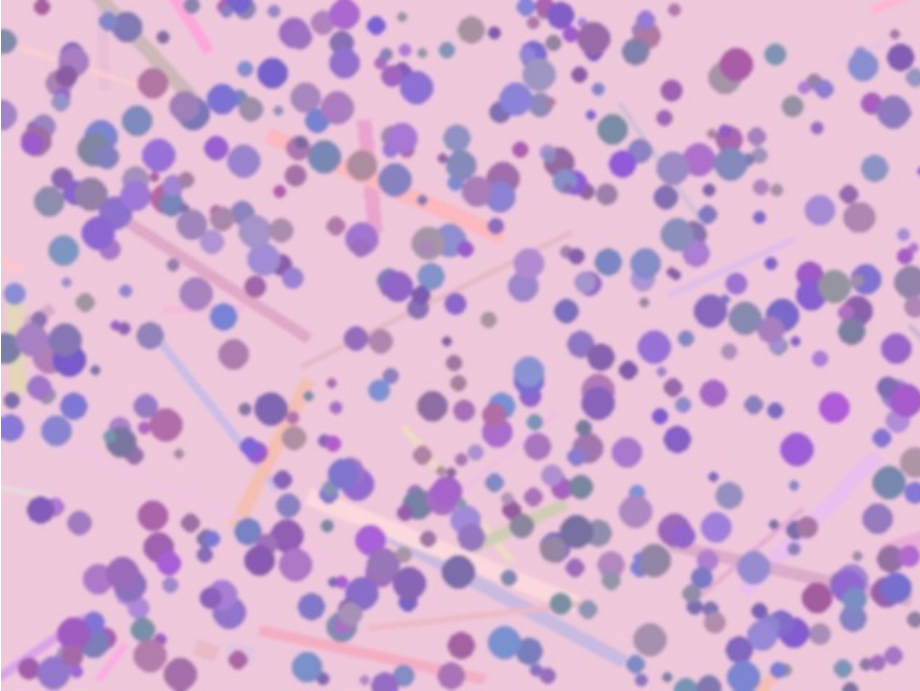


Figure 3: Wet (left) and dry (right) gangrene

Wet gangrene shows a swollen, moist limb with a poorly defined line of demarcation. Dry gangrene shows a shrunken, black, mummified limb with a clear line of demarcation.



Thyroid carcinoma



Papillary thyroid carcinoma

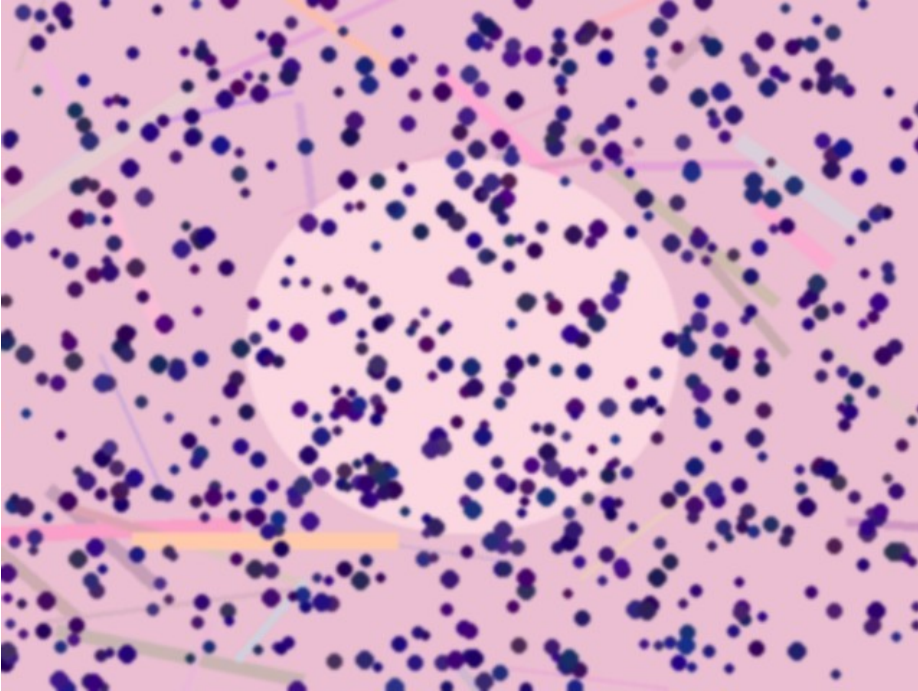


Follicular thyroid carcinoma

Papillary carcinoma shows psammoma bodies and nuclei with a ground-glass appearance. Follicular carcinoma is diagnosed by capsular or vascular invasion.



Diabetes mellitus: the islets



Type 1 diabetes: insulitis with lymphocytes



Type 2 diabetes: amyloid deposition in an islet

Type 1 diabetes mellitus results from autoimmune destruction of pancreatic beta cells. Type 2 diabetes mellitus is characterized by insulin resistance and relative insulin deficiency.



Normal and cirrhotic liver



Normal liver

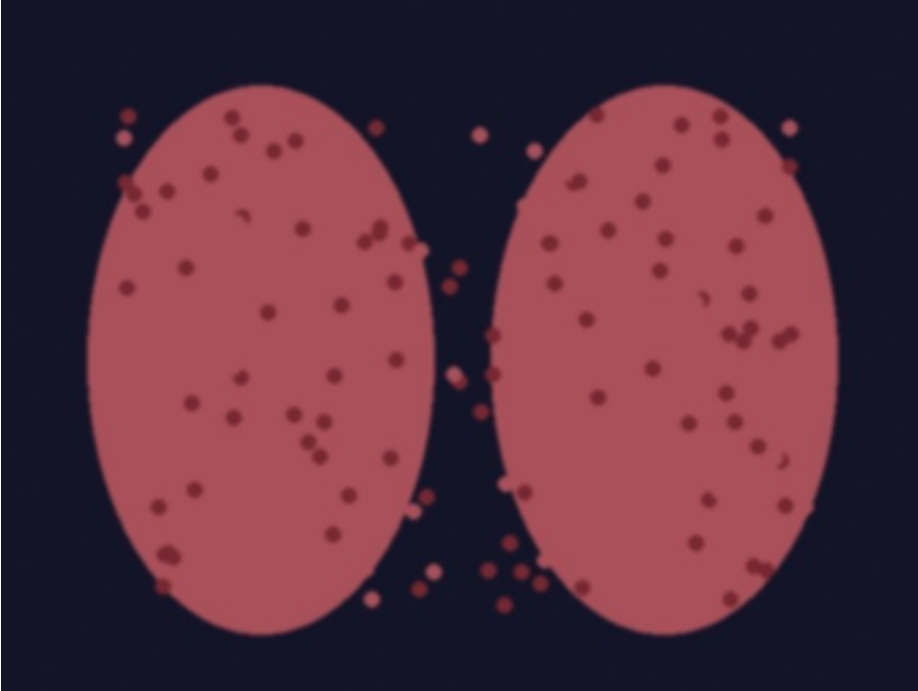


Cirrhotic liver with regenerative nodules

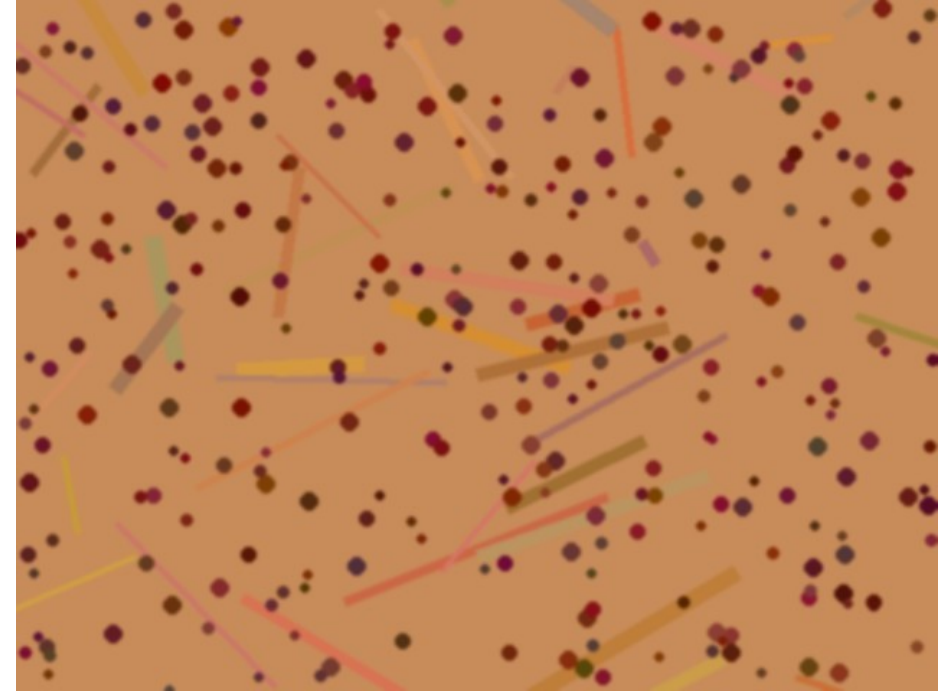
Cirrhosis is characterized by bridging fibrous septa and regenerative parenchymal nodules.



Heart failure



Left-sided heart failure: pulmonary edema

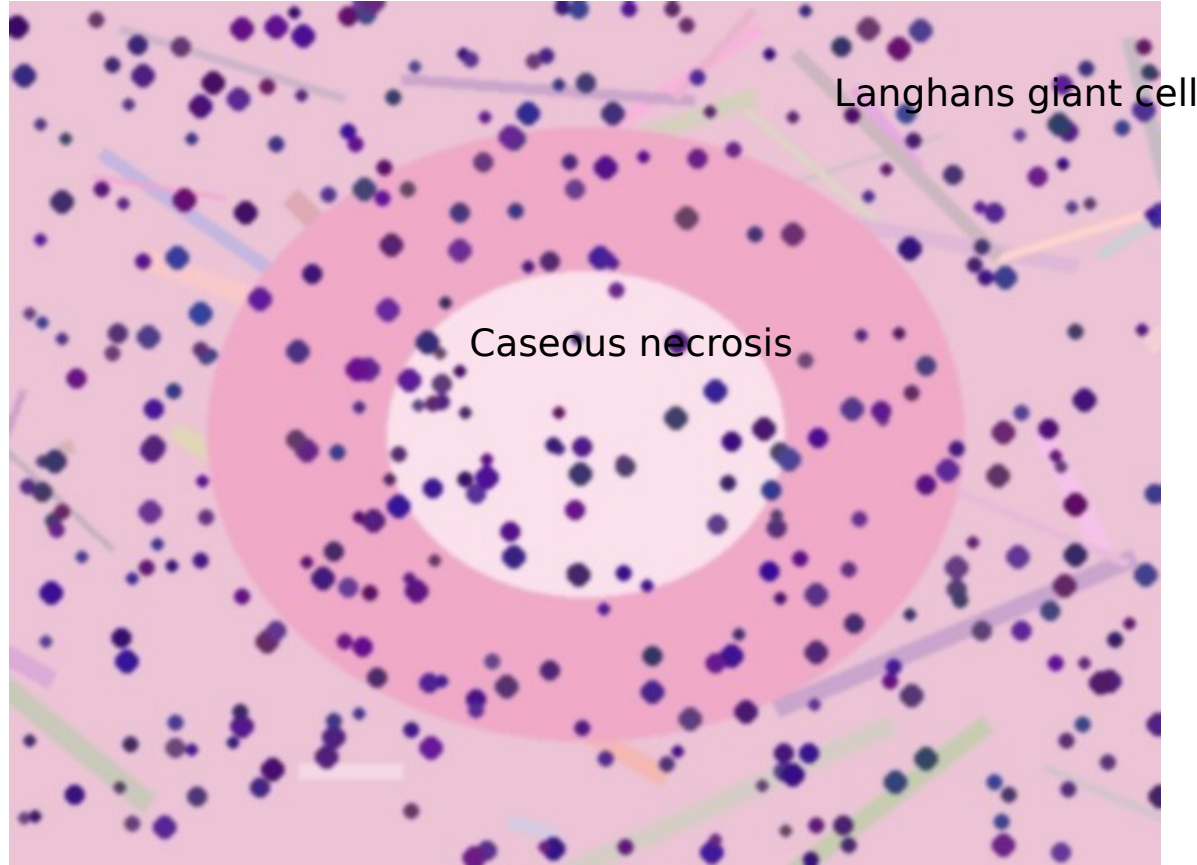


Right-sided heart failure: nutmeg liver

Left-sided heart failure most commonly causes pulmonary congestion and edema. Right-sided heart failure is most often caused by left-sided heart failure. It causes congestive hepatomegaly with a nutmeg appearance.



Tuberculous granuloma



A tuberculous granuloma consists of central caseous necrosis surrounded by epithelioid macrophages and Langhans giant cells.

Figure 7: Tuberculous granuloma



Amyloidosis

Type	Protein	Association
Primary (AL)	Light chains	Plasma cell dyscrasia
Secondary (AA)	Serum amyloid A	Chronic inflammation
Senile	Transthyretin	Aging heart
Dialysis	β 2-microglobulin	Long-term dialysis

Table 2: Classification of amyloidosis



Congo red stain under polarized light

Amyloid shows apple-green birefringence under polarized light after Congo red staining.



Fatty change



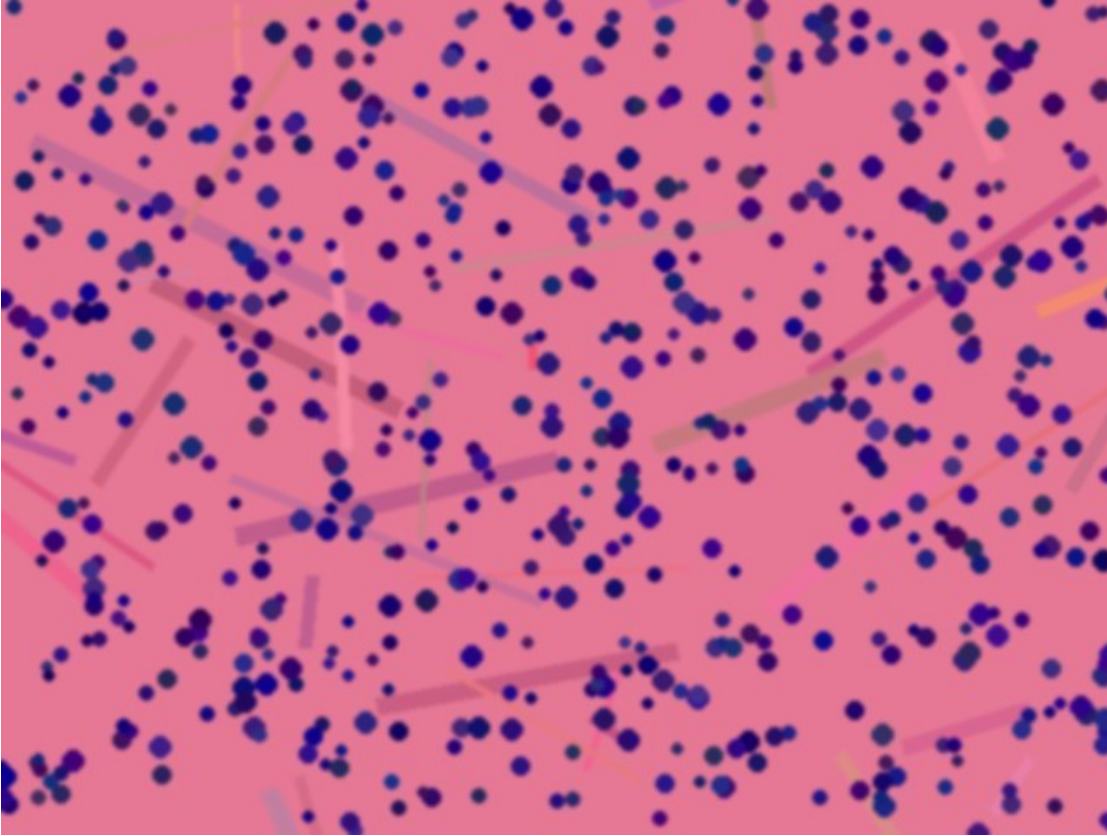
Hepatic steatosis is most often caused by alcohol abuse and nonalcoholic fatty liver disease. Lipid accumulates as clear vacuoles in the cytoplasm of hepatocytes.

Figure 8: Hepatic steatosis





Myocardial infarction

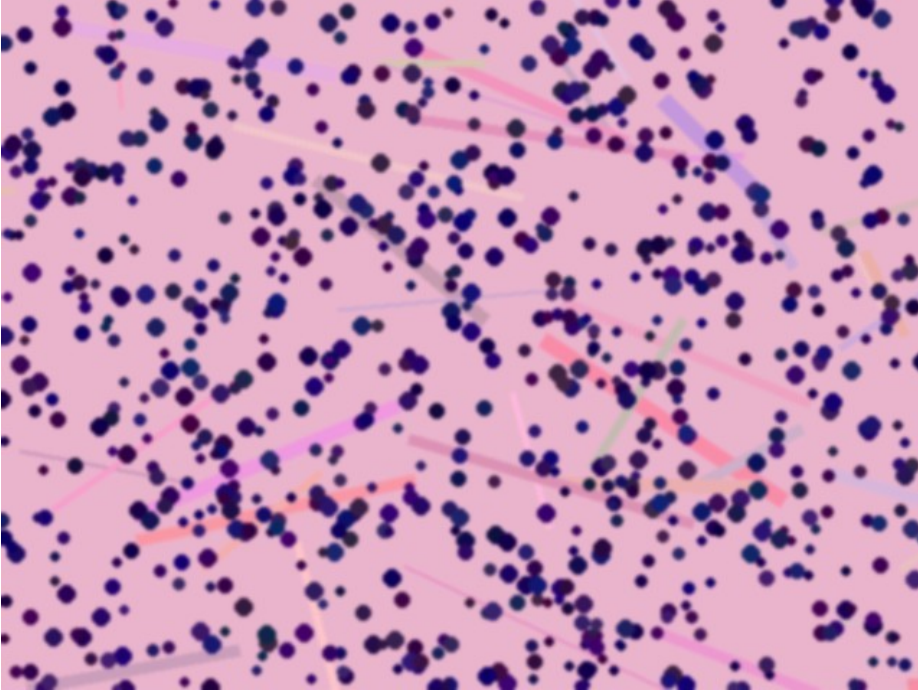


Neutrophil infiltration peaks 1 to 3 days after myocardial infarction.

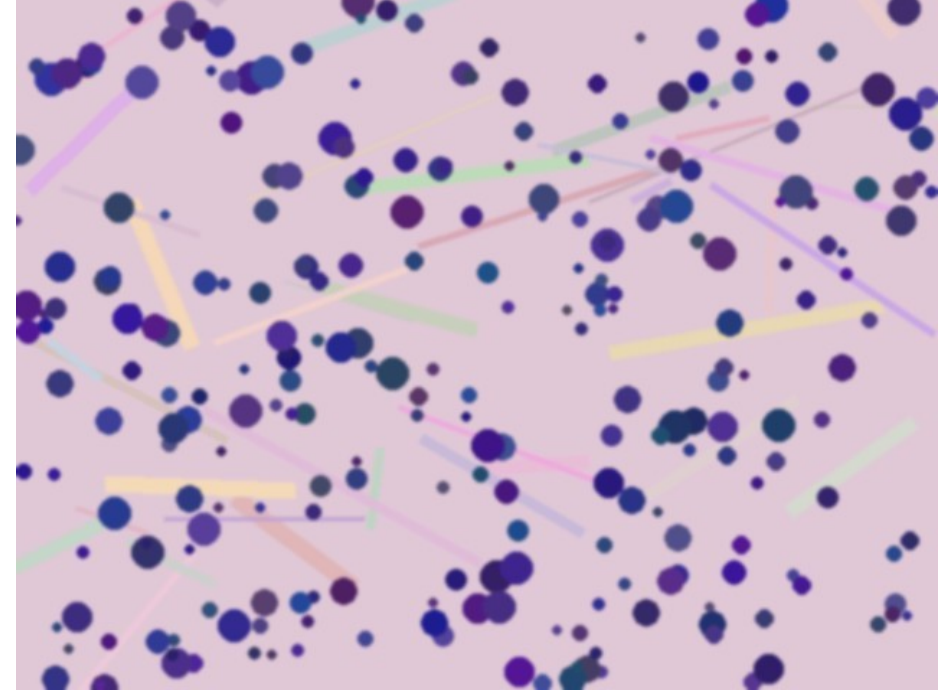
Macrophages remove dead myocytes 3 to 7 days after infarction.

Figure 9: Myocardial infarction at 3 days with neutrophils

Acute and chronic inflammation



Acute appendicitis with neutrophils



Chronic cholecystitis with lymphocytes

Acute inflammation is dominated by neutrophils. Chronic inflammation is characterized by lymphocytes, plasma cells and macrophages.

End of image review

Thank you